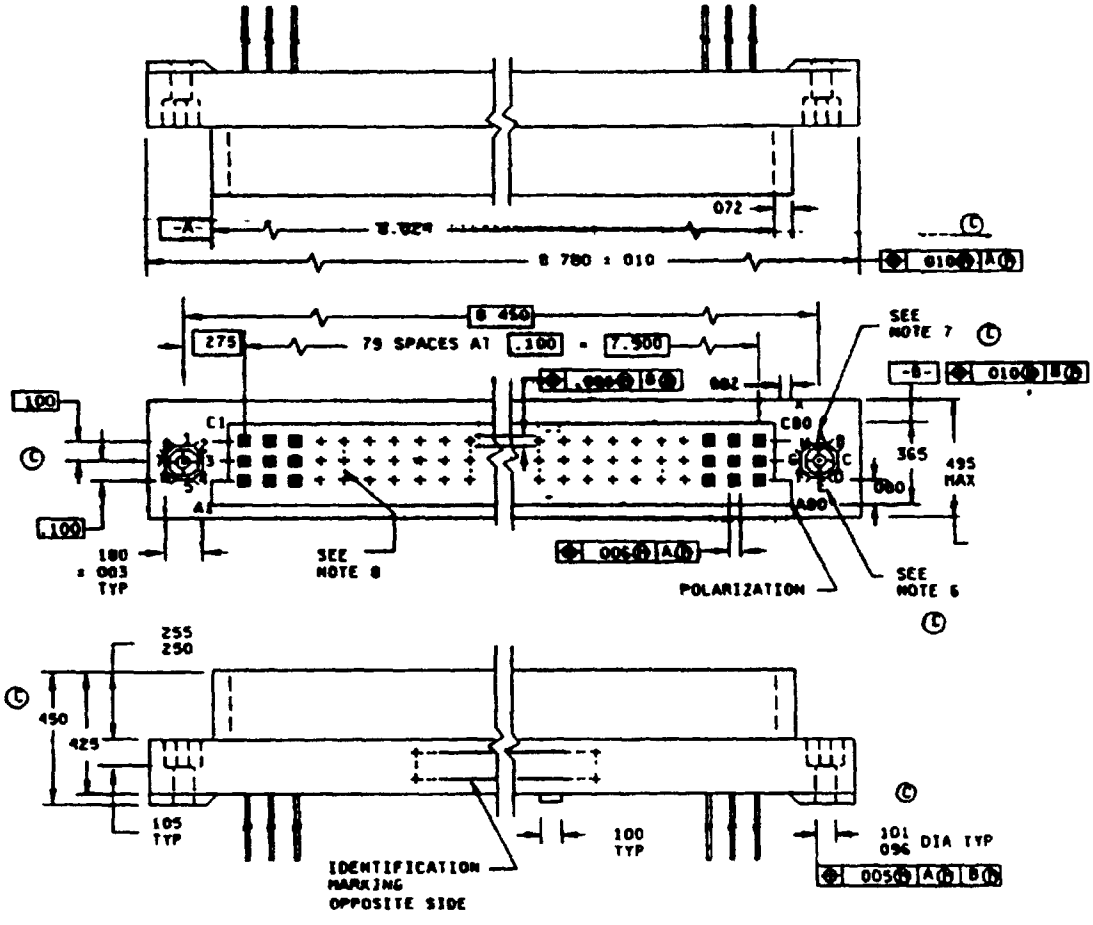
INCH-POUND
MIL-C-55302/130C(EC)
20 April 1993
SUPERSEDING
MIL-C-55302/130B(EC)
18 August 1988

MILITARY SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES RECEPTACLES, 90 THROUGH 240 CONTACT POSITIONS, FOR PRINTED WIRING BOARDS (.100 INCH SPACING)

This specification sheet is approved for use by the Space and Naval Warfare Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-55302.

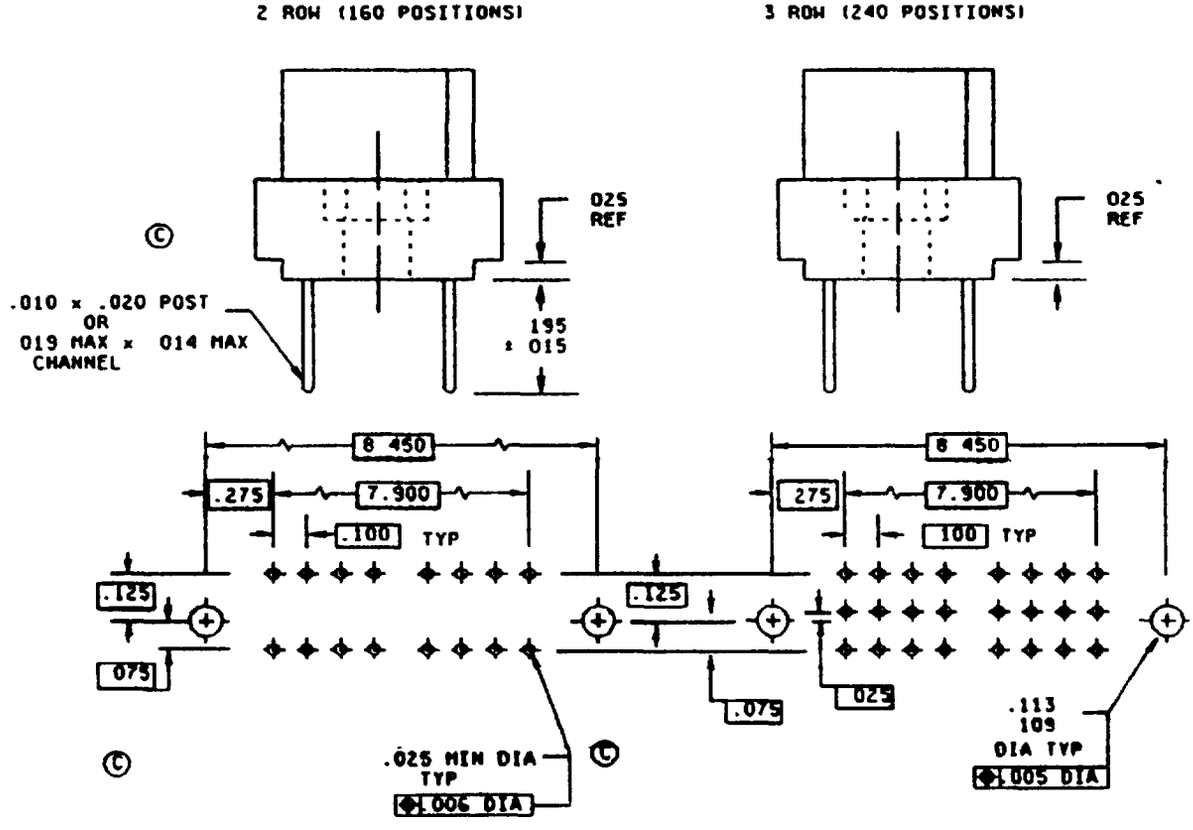


© FIGURE 1 Connectors, kit plug, 100 (2.54 mm) spacing, 160 and 240 contacts

© denotes changes.

DISTRIBUTION STATEMENT A Approved for public release, distribution is unlimited

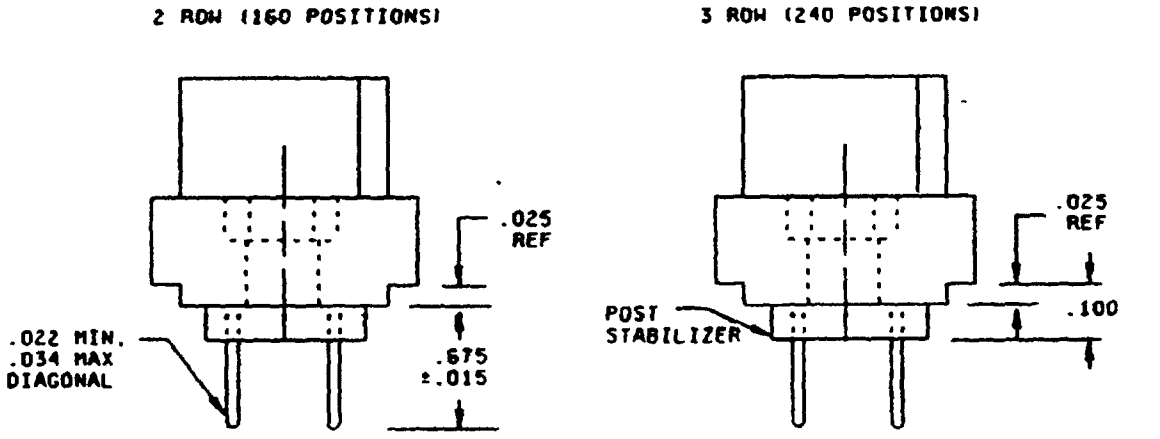
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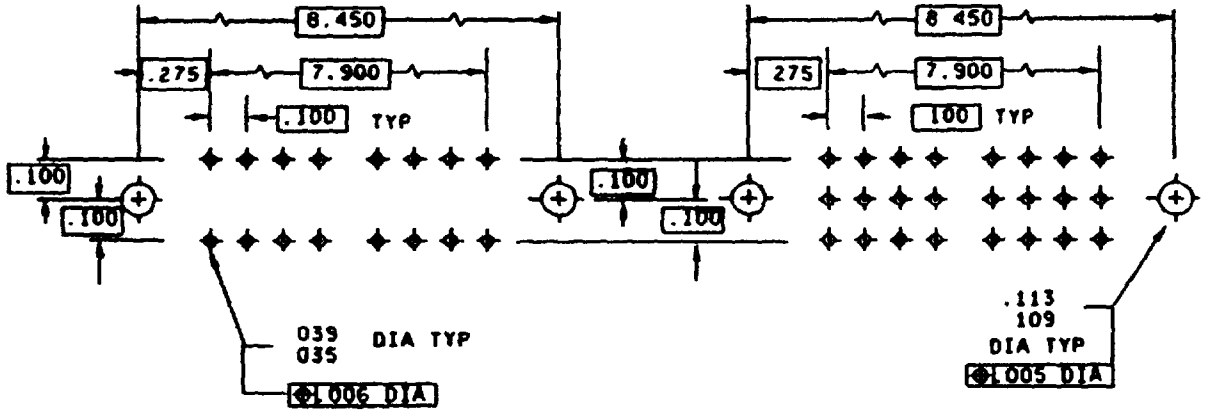
Contact types and recommended circuit layouts

© FIGURE 1 Connectors, kit plug, .100 (2.54 mm) spacing, 160 and 240 contacts - Continued

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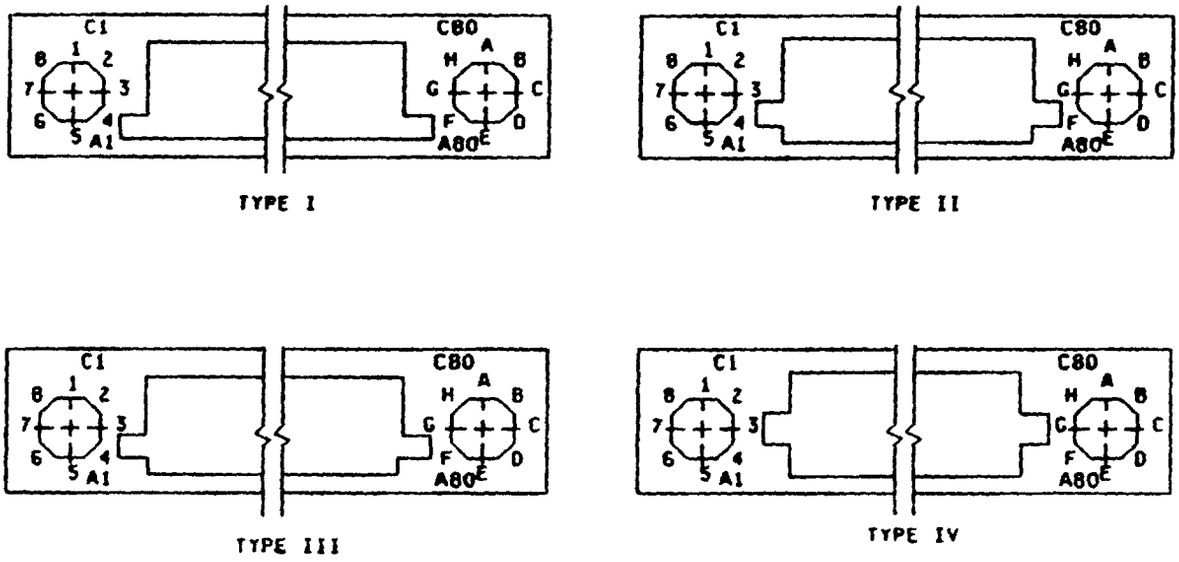
CONTACT TYPE 02



Contact types and recommended circuit layouts

© FIGURE 1 Connectors, kit plug, .100 (2.54 mm) spacing, 160 and 240 contacts - Continued

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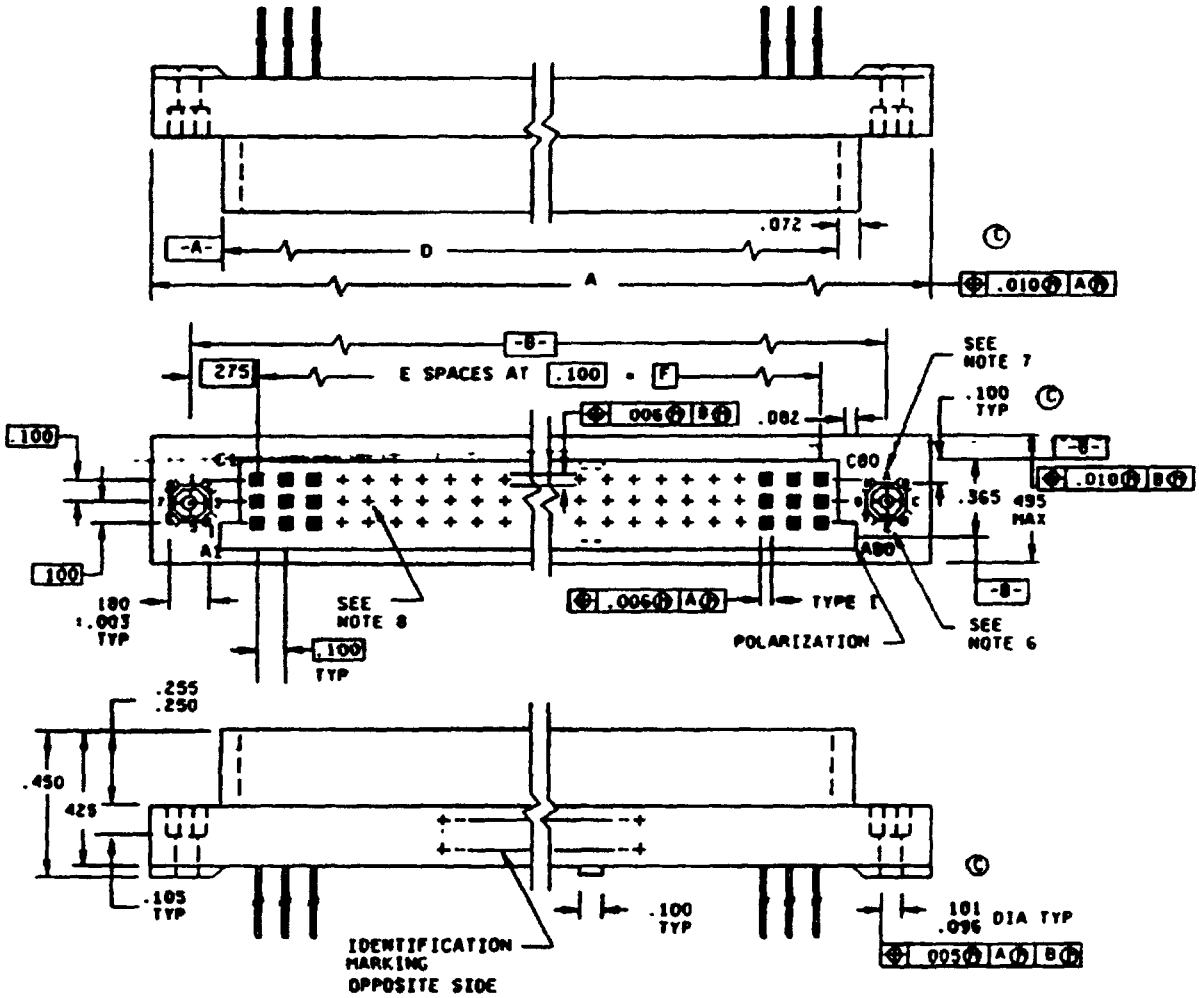


Inches	mm	Inches	mm	Inches	mm
003	0.08	.039	0.99	.225	6 48
.005	0 13	.072	1.83	.275	6.99
.006	0.15	.073	1.85	.365	9 27
.010	0.25	.080	2.03	.425	10.80
014	0.36	.096	2 44	450	11.43
.015	0.38	100	2.54	495	12.57
019	0.48	109	2 77	675	17.15
.022	0 56	.113	2.87	7 700	200.66
.027	0.69	180	4 57	8.024	203.81
034	0 86	.195	4.95	8 450	214 63
.035	0.89	250	6.35	8 750	222 25

NOTES
1. Dimensions are in inches.
2. Metric equivalents are given for general information only
3. Unless otherwise specified, tolerance is ±.005 (0.13 mm) on decimals and +2° on angles.
4. These connectors mate with connectors specified in MIL-C-55302/129
5. Use printed circuit boards with both the type 1 and type 2 receptacles
6. (C) Numbers indicating end cavities and letters indicating rows marked or molded this surface
7. Location indicators embossed on surface
8. (C) Lines indicate every five cavities marked or molded on mating face except end cavities

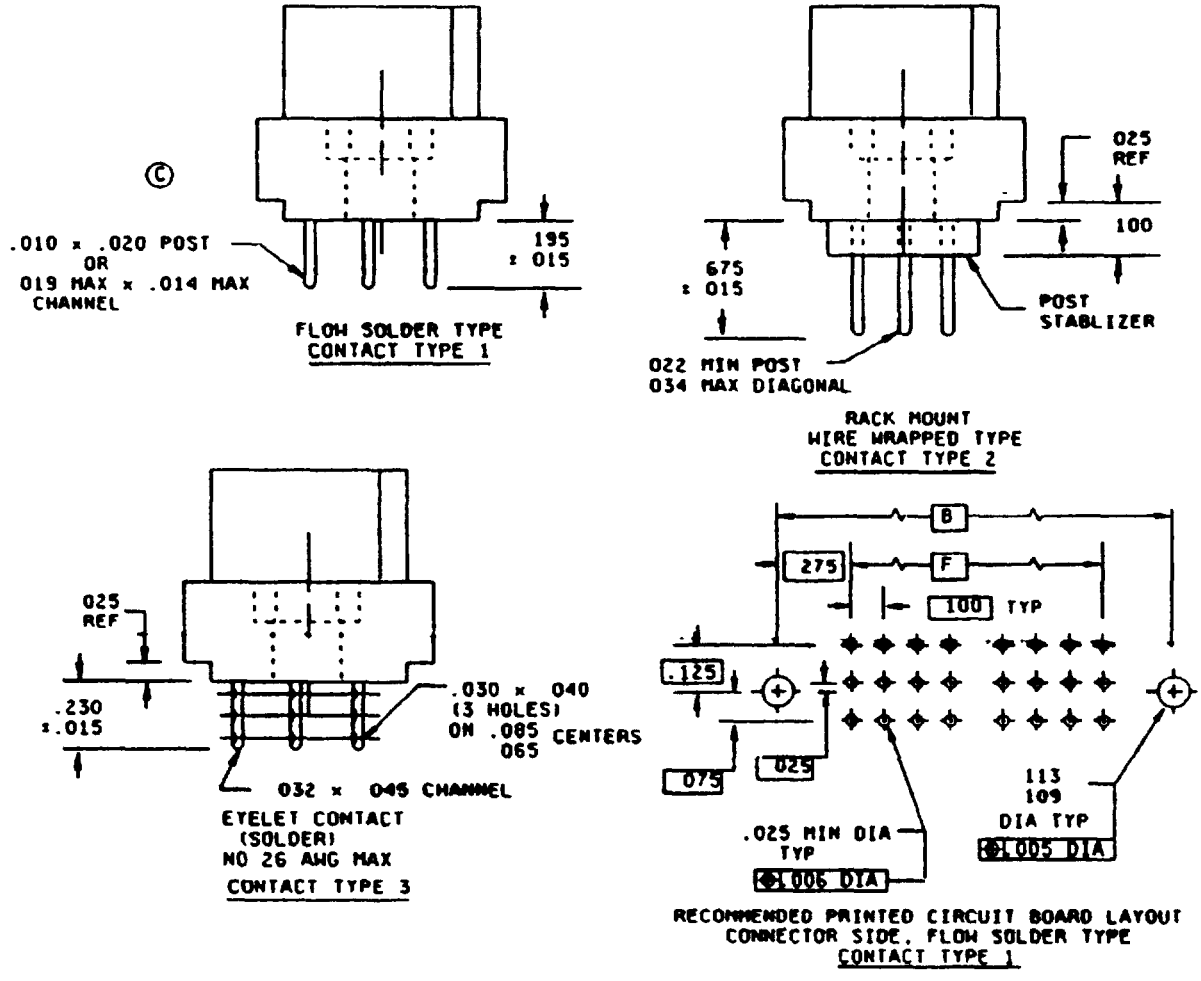
(C) FIGURE 1 Connectors, kit plug, 100 (2.54 mm) spacing, 160 and 240 contacts - Continued

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© FIGURE 2. Connectors, receptacles, 100 (2.54 mm) spacing, 90 through 240 contact positions

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Inches	mm	Inches	mm	Inches	mm
.005	0.13	027	0 69	085	2 16
006	0 15	030	0 76	100	2 54
.014	0 36	.032	0 81	125	3 18
015	0.38	034	0 86	195	4 95
019	0 48	040	1 02	675	17 14
022	0 56	045	1 14		
025	0 64	075	1 90		

NOTES
1 Dimensions are in inches
2 Metric equivalents are given for general information only

© FIGURE 2 Connectors, receptacles, .100 (2.54 mm) spacing, 90 through 240 contact positions - Continued

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REQUIREMENTS.

Design and construction:

Dimensions and configurations: See figures 1 and 2, and tables I and II

(C) Insulator body: In accordance with MIL-C-55302 or MIL-M-24519 type GLCP-30F

(C) Contact material: Contact material shall be in accordance with MIL-C-55302 or beryllium copper alloy CDA174 in accordance with ASTM B768.

Plating: Types 1, 2, and 3 shall be nickel plated per MIL-C-55302 Over the nickel plate, the mating engagement area of types 1 and 3 for a minimum length of 100 inch and the entire contact type 2 shall be gold plated per MIL-C-55302. Over the nickel plate, the solder tail area for types 1 and 3 shall be tin-lead plated for .215 and .245 inch, respectively per MIL-C-55302

Contact identification. See figures 1 and 2, and tables I and II

Current rating: 3.0 amperes, maximum per contact, 2.25 amperes continuous per contact at 75°F.

Keying (see MIL-C-55302/31).

Dash numbers -01 through -16. Two keys, part number M55302/31-08, and two key mounting hardware, part number M55302/31-09, packaged with connectors.

Dash numbers -17 through -46: Key, part number M55302/31-04, with 086-56 UNC-2A mounting screw is recommended

Replacement contacts: See MIL-C-55302/32.

Contact removal: See MIL-I-81969/9 for contact types 1 and 3 Use Amp 1-265871-7 tool for contact type 2.

Mating and unmating: The maximum mating force in pounds shall be the number of contacts multiplied by 0.25, the withdrawal force in pounds shall be a minimum of 0 025 times the number of contacts and shall not exceed the measured insertion force

Contact resistance: No individual contact pair shall have a resistance exceeding 0.020 ohm

Contact retention: 3 pounds, minimum

Dielectric withstanding voltage:

Sea level: 900 volts rms.

High altitude. 200 volts rms.

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Resistance to test probe damage The connector shall meet the requirements of MIL-C-55302/30

Contact separation force The connector shall meet the requirements of MIL-C-55302/29

Oversized pin exclusion 0465 diameter pin

Part number M55302/130- (and dash number from table I for 160 and 240 contact positions).

M55302/130- (and dash number from table II for 90 through 240 contact positions)

Patent number 3,404,367 The Government has a royalty free license under this patent for the benefit of manufacturers of the item either for the Government or for use in equipment to be delivered to the Government

TABLE I Dash number; polarization type, contact type for 160 and 240 contact positions.

Dash number	Number of contacts	Polarization type	Contact type
01	160	I	1
02	160	I	2
03	160	II	1
04	160	II	2
05	160	III	1
06	160	III	2
07	160	IV	1
08	160	IV	2
09	240	I	1
10	240	I	2
11	240	II	1
12	240	II	2
13	240	III	1
14	240	III	2
15	240	IV	1
16	240	IV	2

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(C) TABLE II Dash numbers and dimensions for 90 through 240 contact positions

Dash no.	No of contacts	(C) Contact type	A	B	D	E	F	Contact ident no	(C) Polarization type
17 18 19	90	1 2 3	3.750 (95.25)	3 450 (87.63)	3.024 (76.81)	29	2 900 (73.66)	1-30	I
20 21 22	105	1 2 3	4 250 (107.95)	3.950 (100.33)	3.524 (89 51)	34	3 400 (86 36)	1-35	I
23 24 25	120	1 2 3	4 750 (120.65)	4.450 (113.03)	4.024 (102.21)	39	3 900 (99 06)	1-40	I
26 27 28	135	1 2 3	5 250 (133 35)	4.950 (125.73)	4 524 (115.57)	44	4 400 (111 76)	1-45	I
29 30 31	150	1 2 3	5 750 (146 05)	5.450 (138 43)	5 024 (127 61)	49	4 400 (124 46)	1-50	I
32 33 34	165	1 2 3	6 250 (158 75)	5 950 (151.13)	5 524 (140.31)	54	5.400 (137 16)	1-55	I
35 36 37	180	1 2 3	6.750 (171 45)	6 450 (163 83)	6 024 (153.01)	59	5 900 (149 86)	1-60	I
38 39 40	195	1 2 3	7 250 (184.15)	6 950 (176 53)	6.524 (166.37)	64	6 400 (162 56)	1-65	I
41 42 43	210	1 2 3	7.750 (196.85)	7 450 (189.23)	7 024 (178.41)	69	6 900 (175 26)	1-70	I
44 45 46	240	1 2 3	8.750 (222.25)	8.450 (214.63)	8 024 (203.81)	79	7 900 (200 66)	1-80	I

CONCLUDING MATERIAL

Custodian
Navy - EC

Preparing activity
Navy - EC

Agent
DLA - ES

(Project 5935-N488-02)

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